MODULE 4 MALE REPRODUCTIVE SYSTEM

The Male Reproductive System.

THE REPRODUCTIVE SYSTEM, MALE

A reference for the male reproductive system video and lab. This page covers the scrotum and testes, the duct system and the path of sperm, the structure of a sperm cell, the accessory glands, and the penis. The focus is on the structures and the job each one does.

BY THE END

1. Identify the structures of the scrotum and the testes.
2. Trace the path of sperm through the duct system.
3. Name the accessory glands and the parts of a sperm cell and the penis.

WATCH AFTER YOU READ THIS CHAPTER

Concept videos: The male reproductive system

drsrennie-stack.github.io/new-build-bio4-solano/male-repro-concept-videos.html

The Male Reproductive System, an Overview

The male reproductive system produces sperm and delivers it. Its parts sort into the gonads, a duct system, the accessory glands, and the supporting structures.

THE FOUR GROUPS OF PARTS

Structure	What it is
Testes	The male gonads; they produce sperm and secrete hormones
Duct system	The epididymis, ductus deferens, ejaculatory ducts, and urethra, which store, mature, and transport sperm
Accessory glands	The seminal vesicles, prostate, and bulbourethral glands, which add fluid to form semen
Supporting structures	The scrotum and the penis, which help deliver sperm into the female tract

The Scrotum and Testes

The testes hang outside the body in the scrotum, which keeps them a few degrees cooler than the core. Inside each testis, coiled tubules produce the sperm.

THE SCROTUM OUTSIDE AND THE TESTIS WITHIN

Structure	What it is
Scrotum	The pouch of skin and subcutaneous tissue that supports the testes; a raphe marks its surface and a scrotal septum divides it within
Dartos muscle	The smooth muscle of the scrotum that wrinkles the skin to reduce heat loss
Cremaster muscle	The skeletal muscle that raises and lowers the testes to regulate their temperature
Tunica vaginalis	The serous membrane, derived from the peritoneum, that partly covers each testis
Tunica albuginea	The white fibrous capsule that divides each testis into two hundred to three hundred lobules
Seminiferous tubules	The tightly coiled tubes within the lobules where sperm are produced
Sustentacular cells	The Sertoli cells lining the seminiferous tubules; they nourish developing sperm and form the blood-testis barrier
Interstitial cells	The Leydig cells between the tubules; they produce testosterone

HOLD ONTO THIS

Two muscles set the temperature of the testes, and they are different tissues. The dartos is smooth muscle that wrinkles the scrotal skin; the cremaster is skeletal muscle that raises and lowers the testes themselves.

The Duct System and the Path of Sperm

After they form, sperm travel a fixed series of ducts that store them, mature them, and carry them out.

THE DUCTS, IN ORDER, AND THE CORD THAT CARRIES THEM

Structure	What it is
Epididymis	The comma-shaped, coiled duct on the testis where sperm are stored and gain motility
Ductus deferens	The vas deferens; the duct that carries sperm from the epididymis up toward the urethra
Ejaculatory duct	The short duct formed where the ductus deferens meets the seminal vesicle duct; it empties into the prostatic urethra
Urethra	The shared final duct; it passes through the prostate, the perineum, and the penis
Spermatic cord	The bundle that contains the ductus deferens, blood vessels, nerves, and the cremaster muscle

Sperm follow one route from the testis to the outside.

1. **Seminiferous tubules** sperm are produced inside the testis
2. **Epididymis** sperm are stored and gain the ability to move
3. **Ductus deferens** sperm are carried up out of the scrotum

4. **Ejaculatory duct** the ductus deferens joins the seminal vesicle duct
5. **Urethra** sperm pass through the prostatic, membranous, and spongy urethra to the outside

HOLD ONTO THIS

Only the seminiferous tubules make sperm. Every other structure on this route stores, matures, or carries them, so sort the male structures into one that produces and the rest that transport.

The Structure of a Sperm Cell

A mature sperm cell is built for one job, reaching and entering an oocyte. It has a head and a tail.

THE HEAD AND THE PARTS OF THE TAIL

Structure	What it is
Head	The part of the sperm that holds the nucleus with its twenty three chromosomes
Acrosome	The enzyme-filled cap over the head that helps the sperm penetrate the oocyte
Neck	The part of the tail just behind the head, containing the centrioles
Middle piece	The part of the tail packed with mitochondria, which power movement
Principal piece	The longest part of the tail
End piece	The tapering terminal part of the tail

The Accessory Glands and Semen

Three sets of accessory glands add fluid to the sperm. The mixture of sperm and these fluids is semen.

COMPARE THE GLANDS

Gland	What it adds
Seminal vesicles	An alkaline fluid rich in fructose that nourishes the sperm; the largest share of semen
Prostate	A milky, slightly acidic fluid that helps the sperm stay active
Bulbourethral glands	The Cowper's glands; an alkaline mucus that neutralizes urine in the urethra and lubricates

AND THE MIXTURE ITSELF

Structure	What it is
Semen	The mixture of sperm and the fluids of the three accessory glands

The Penis

The penis delivers semen and carries the urethra. It is built around three columns of erectile tissue.

THE REGIONS OF THE PENIS AND ITS ERECTILE COLUMNS

Structure	What it is
Root	The attached portion of the penis, containing the bulb and the crura
Body	The free, movable portion of the penis
Glans penis	The enlarged, acorn-shaped tip, an extension of the corpus spongiosum
Prepuce	The foreskin; the loose fold of skin over the glans
Corpora cavernosa	The two dorsal columns of erectile tissue
Corpus spongiosum	The single ventral column of erectile tissue that surrounds the spongy urethra
Erectile tissue	The spongy tissue of the columns that fills with blood during an erection

Common Disorders of the Male Reproductive System

Compare the common disorders of the male reproductive system by the structure each one affects.

COMMON DISORDERS OF THE MALE REPRODUCTIVE SYSTEM

Disorder	What it is
Cryptorchidism	One or both testes fail to descend into the scrotum
Hydrocele	A collection of serous fluid in the tunica vaginalis around a testis
Varicocele	An abnormal swelling of the veins that drain the testis
Benign prostatic hyperplasia	A non-cancerous enlargement of the prostate that can squeeze the urethra
Prostate cancer	A malignant tumor of the prostate gland
Testicular cancer	A malignant tumor of the testis
Inguinal hernia	A loop of intestine pushes through a weak point in the abdominal wall at the inguinal canal

WHY THE TESTES HANG OUTSIDE

Sperm production needs a temperature a few degrees below the body core, which is why the testes sit in the **scrotum** and the **cremaster** and **dartos** muscles adjust their position. In **cryptorchidism**, an undescended testis stays at body temperature, and if it is not corrected the warmth can destroy the sperm-forming cells.

The duct system shares its final stretch with the urinary tract. Because the **urethra** passes straight through the **prostate**, an enlarged prostate squeezes the urethra and makes urination difficult, which is how a reproductive organ produces a urinary symptom.